

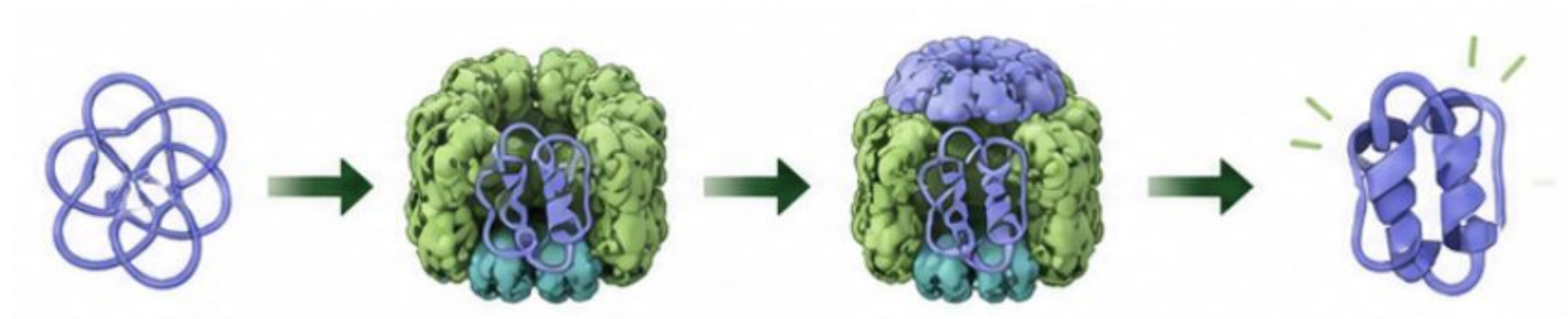
Sequence Similarity, Alignment, and Phylogenetic Inference of GroEL Gene in Bacteria

Rebeca Velis Ventura

Background

❖ GroL Gene

- Heat shock protein
- Assist in protein folding and maintain protein stability
- To investigate the evolutionary relationships among other bacteria
- Compare GroEL sequences of different bacteria
- Conserved regions of sequences



Methods

BLAST Analysis *

- ❖ FAST sequence
- ❖ NCBI BLAST
 - blastp
- ❖ Source Organism
 - *Escherichia coli*
- ❖ Target Organism
 - *Thermus thermophilus*

Multiple Sequence Alignment *

- ❖ Aligner
 - Clustal Omega
 - DNA
 - Protein

Phylogenetic Analysis *

- ❖ Software use
 - Simple Phylogeny
 - ITOL

Results

- ❖ BLAST Analysis
 - Significant GroEL matches identified
 - 97% query coverage
 - E-value is 0.0, alignment is significant

MULTISPECIES: chaperonin GroEL [Thermus]
Sequence ID: [WP_008630425.1](#) Length: 543 Number of Matches: 1
[See 1 more title\(s\)](#) [See all Identical Proteins\(IPG\)](#)

Range 1: 2 to 537 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
645 bits(1664)	0.0	Compositional matrix adjust.	337/537(63%)	418/537(77%)	4/537(0%)
Query	3	AKDVKFGNDARVKMLRGVNVLADAVKVTLGPKGRNVVLDKSFGAPTITKDGVSVAREIEL	62		
Sbjct	2	AK + F AR + RGVN +A+AVKVTLG+GRNVVL+K FG+PTITKDG+VA+E+EL	61		
Query	63	AKILVFDEAARRALERGVNAVAVKVTLGPRGRNVVLEKKFGSPTITKDGVTVAKEVEL	61		
Query	63	EDKFENMGAQMVKEVASKANDAAGDGTATVLAQAIIITEGLKAVAAGMNPMDLKRIGDK	122		
Sbjct	62	ED EN+GAQ++KEVASK ND AGDGTATVLAQAII+ EGLK VAAG NP+ LKRG+K	121		
Query	123	EDHLENIGAQLLKEVASKTNDVAGDGTATVLAQAIVREGLKNVAAGANPLALKRGIEK	121		
Query	123	AVTAAVEELKALSVPCSDSKAIAQVGTISANSDETGVGLIAEAMDKVGKEGVITVEDGTG	182		
Sbjct	122	AV AAVE++KAL++P D KAI +V TISAN D VGKLIA+AM+KVGKEG+ITVE+	180		
		AVEAAVEKIKALAIPVEDRKAIEEVATISAN-DPEVGKLIADAMEKVGKEGIITVEESKS	180		

Sequences producing significant alignments				Download		Select columns		Show	100	
☒ select all 7 sequences selected				GenPept	Graphics	Distance tree of results	Multiple alignment	MSA Viewer		
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	
☒	MULTISPECIES: chaperonin GroEL [Thermus]	Thermus	645	645	97%	0.0	62.76%	543	WP_008630425.1	
☒	MULTISPECIES: chaperonin GroEL [Thermus]	Thermus	644	644	97%	0.0	62.57%	543	WP_011174077.1	
☒	chaperonin GroEL [Thermus thermophilus]	Thermus thermo...	644	644	97%	0.0	62.38%	543	BHR94817.1	
☒	chaperonin GroEL [Thermus thermophilus]	Thermus thermo...	644	644	97%	0.0	62.57%	543	WP_096411521.1	
☒	chaperonin-60 [Thermus thermophilus HB27]	Thermus thermo...	641	641	97%	0.0	62.38%	543	CAB65482.1	
☒	RecName: Full=Chaperonin GroEL; AltName: Full=60 kDa chaperonin; AltName: Full=Chaperonin-60; Short=C...	Thermus thermo...	207	207	26%	1e-64	69.44%	145	P61491.2	
☒	cpn60=58 kda chaperonin [Thermus thermophilus, Peptide Partial, 43 aa].[Thermus thermophilus]	Thermus thermo...	50.4	50.4	8%	2e-08	60.47%	43	AAB20363.1	

Results

❖ Multiple Sequence Alignment

➤ Clustal Omega

➤ Protein

➤ DNA

	1	10	20	30	40	50
Corynebacterium glutamicum groL	.MAKLI	AFDQDAREGILR	GVDALANA	VKVTLGPRGRNVVLDKA	FGGPLVTNDGV	TIARDI
Bifidobacterium animalis groL	.MAKII	EYDEEARQGMLA	GLDKLADT	VKVTLGPKGRNVVLDKT	YGAPTITNDGV	SIAKEI
Escherichia coli groL	MAAKD	VKFGNDARVKMLR	GVNVLADAVKVTLGPKGRNVVLDKS	FGAPTITKDGV	SVAREI	
Thermus thermophilus groL	.MAKIL	VFDEAARRALER	GVNAVANA	VKVTLGPRGRNVVLEKK	FGSPTITKDGV	TVAKEV

	60	70	80	90	100	110	
Corynebacterium glutamicum groL	DLEDPFENL	GAQLVKSVA	VKTNDI	AGDGTTTATL	LAQALIAEGLRN	VAAAGANPME	LNKGI
Bifidobacterium animalis groL	DLEDPFERI	GAELVKEVA	KRTDDV	AGDGTTTATV	LAQSLVHEGLKN	VVAGSNPIAL	RRGI
Escherichia coli groL	ELEDKFENM	GAQMVKEVA	SKANDA	AGDGTTTATV	LAQAIIITEGLKA	VAAAGMNPMD	LKRGI
Thermus thermophilus groL	ELEDHLENI	GAQLLKEVA	SKTNDV	AGDGTTTATV	LAQAIVREGLKN	VAAAGANPLAL	LKRGI

	1	10	20	30	40	50		
Corynebacterium glutamicum groL	...ATGGCA	AAGCTCAT	TGCTTTGACC	AGGACGCC	CGCGAAGG	GCATTCTC	CGGGGC	GTT
Escherichia coli groL	ATGGCAG	CTAAAGAC	GTAAATTC	GGTAACGAC	GCTCGTGT	GAAAATGCT	GCGCGG	CGTA
Thermus thermophilus groL	...ATGGCG	AAGATCCT	GTGTGTTG	ACGAGGCT	GCGCGCCGC	GCCCTGGAG	CGCGGG	GTC
Bifidobacterium animalis groL	...ATGGCA	AAGATCAT	TGAATAC	GATGAGGA	AGCTCGT	CAGGGTATGCT	CGCCGG	TCTG

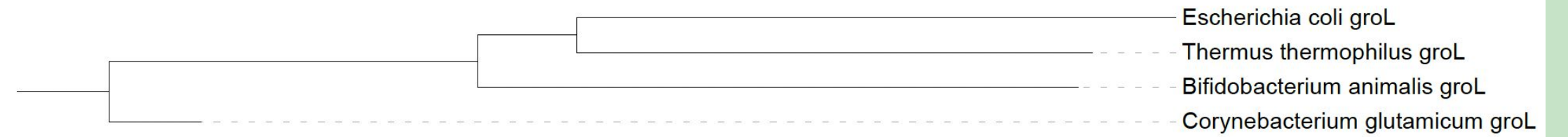
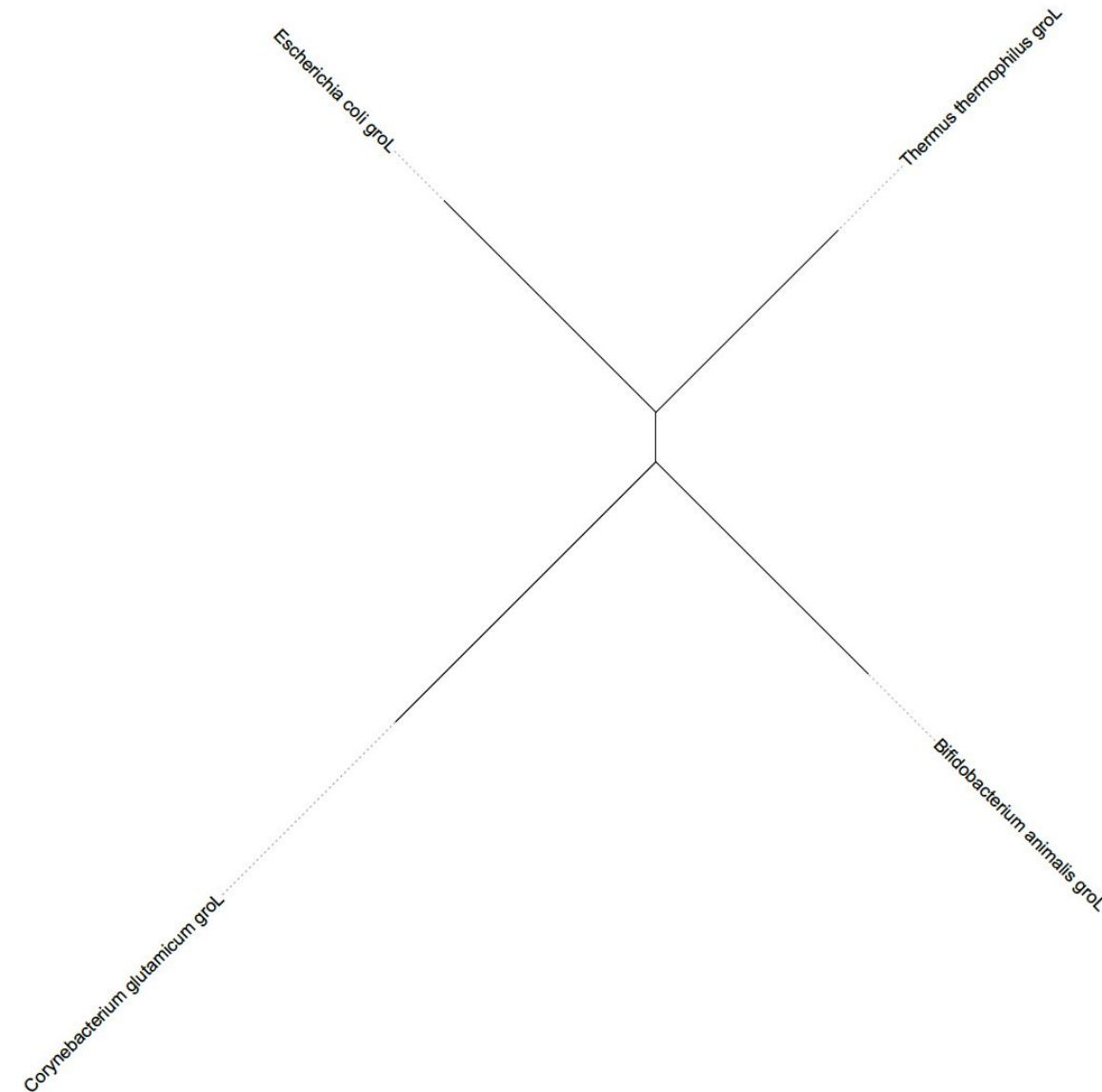
	60	70	80	90	100	110			
Corynebacterium glutamicum groL	GACGCT	CTGGCAAAC	GCTGTCAA	AGGTAAAC	CCTCGGCC	CCACGC	GGCCGT	AACGTG	GTTCTT
Escherichia coli groL	AACGTAC	CTGGCAGAT	GCAGTGAA	AGTTAC	CCTCGGT	CCAAAA	GGCCGT	AACGTAG	GTTCTG
Thermus thermophilus groL	AACGCG	GTTGGCAA	CGCGTGA	AGGTCA	CCTCGGT	CCCTCGG	GGCCGG	AACGTG	GTCCTG
Bifidobacterium animalis groL	GATAAG	CTGGCCGAT	ACCGTCA	AGGTGAC	GCTTGGT	CCCAAG	GGCCGC	AACGTG	GTGCTT

Results

❖ Phylogenetic Analysis

- *E. coli* and *T. thermophilus* show closer relatedness
- *C. glutamicum* is more distantly related
- Shorter branches indicate greater similarity

Tree scale: 0.1





Thank You